PAYROLL MANAGEMENT SYSTEM FOR CONTRACT OF SERVICE FACULTY MEMBERS OF THE CAVITE STATE UNIVERSITY – IMUS CAMPUS USING RFID

Undergraduate Capstone Project
Submitted to the Faculty of the
Department of Computer Studies
Cavite State University-Imus Campus
Imus, Cavite

In partial fulfillment
Of the requirements for the degree of
Bachelor of Science in Information Technology

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October 2026

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An undergraduate thesis manuscript submitted to the faculty of the Department of Computer Studies, Cavite State University - Imus Campus, City of Imus, Cavite, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, with Contribution no. . Prepared under the supervision of Ms. Grace S. Ibañez, MSCS.

INTRODUCTION

Attendance monitoring provides records of faculty members' presence and rendered service hours. For Contract of Service (COS) faculty members, those records support review of the services rendered and provide information used in payroll processing. At Cavite State University - Imus Campus, the Department of Computer Studies (DCS) needs an organized way to record, review, and use attendance and payroll information for its COS faculty members.

The current project is a web-based Payroll Management System connected to an ESP32 attendance device. The software maintains faculty profiles, employment and rank information, work schedules, attendance logs, and payroll records in a centralized database. Faculty attendance can be captured by registered RFID cards or enrolled fingerprints. Authorized users can also record attendance manually when needed.

This chapter describes the implemented system and its intended use for COS faculty members of the DCS. The application includes administrative and faculty accounts, faculty and credential management, attendance monitoring, payroll processing, dashboard summaries, and downloadable payslips. The following sections state its objectives, purpose, scope, and conceptual framework based on the current project files.

The attendance device communicates with the application through its hardware API when a network connection is available. In the active TFT firmware, the ESP32 uses an RFID reader, an AS608 fingerprint sensor, a DS3231 real-time clock, a 3.5-inch ILI9488 display, Wi-Fi, and local flash storage for queued events. If a scan cannot be uploaded, the device retains it for later synchronization with its original event identifier and timestamp. Successful storage on the device does not by itself mean the server has received the event.

The application processes attendance against assigned faculty schedules. It records time-in and time-out, identifies the capture method, and calculates payable hours from overlap with scheduled class blocks after applicable breaks and schedule-based late or undertime rules. Payroll staff can generate a payroll period from attendance, review earnings and deductions, approve records, and release payslips. Faculty users can view their own attendance information and released payslips.

Project Context

The project addresses the need to maintain attendance and payroll information in one accessible system. The implemented application stores records in its database instead of relying on a separate USB retrieval step for each review. The attendance device can send RFID or fingerprint events to the web application through a secured hardware endpoint. When connectivity is interrupted, the TFT device can queue scans locally and retry them after reconnection.

The web application is built with Laravel and a React interface delivered through Inertia. It includes role-based access for administrators, payroll staff, and faculty members. Faculty records can include contact and employment details, faculty rank, service history, assigned schedule blocks and breaks, and registered attendance credentials. RFID cards and fingerprint identifiers are managed separately; fingerprint matching templates are stored by the sensor, while the application keeps the corresponding identifier used to associate a match with a faculty record.

Attendance is recorded as a daily log with time-in and time-out. A registered RFID card or fingerprint can be used for either event. The device and the web application use Philippine time for attendance display and processing. The hardware firmware includes an RTC so scans can still receive a timestamp during a Wi-Fi interruption. Manual attendance entry is also available to authorized payroll staff and is labeled as manual in the log.

Payroll periods follow the system's semi-monthly cutoffs: days 1 to 15 and day 16 to the end of the month. Generated payroll records are drafts. Regular pay is calculated from payable attendance hours and the faculty member's configured hourly or daily rate. Payroll staff can enter applicable earnings and deductions, then approve and release each record. A faculty user can access only their own released payroll record. The system produces printable and downloadable A4 PDF payslips.

Objectives of the Study

The main objective of this study is to develop and refine a payroll management system for COS faculty members of Cavite State University - Imus Campus, specifically those under the Department of Computer Studies. The system combines schedule-based attendance capture and payroll record management through a centralized web application and connected attendance hardware.

Specifically, it aims to:

1. Design the system interface and data structure for faculty profiles, user roles, attendance credentials, schedules, attendance logs, payroll periods, payroll records, and payslips.
2. Develop the web application and ESP32 attendance firmware to support RFID and fingerprint attendance capture, manual attendance entry, centralized storage, schedule-based attendance calculations, and payroll processing.
3. Evaluate the operation of the system components, including account access, faculty and schedule management, credential enrollment, attendance recording, offline event queueing and synchronization, payroll generation, payslip review, and dashboard review.

4. Implement the system for its intended users within the Department of Computer Studies, subject to deployment configuration, hardware availability, network access, and user orientation.
5. Review test results and user feedback, then refine the system through development iterations to address identified issues and the requirements of the study.

Purpose of the Description

The project provides an organized platform for managing faculty profiles, attendance, and payroll-related records for COS faculty members under the DCS. Its current capabilities are described below:

1. Role-based accounts provide separate access for administrators, payroll staff, and faculty. Administrative routes control account and faculty management, while faculty views are limited to the signed-in user's own records.
2. Authorized users can maintain faculty information, employment details, rank and pay rate, service history, work schedules, schedule breaks, and active or inactive status. Administrators can initiate RFID and fingerprint registration or replacement through a time-limited device enrollment request.
3. Registered RFID cards and enrolled fingerprints identify a faculty member at the attendance device. RFID is not the only method: both RFID and fingerprint are implemented attendance methods. An administrator can also enter attendance manually through the web interface.

4. Attendance events are associated with a faculty member, date, time, and capture method. A daily log contains time-in and time-out. The system applies a short time-out cooldown and prevents attendance capture on a day without an assigned schedule.
5. The attendance calculator compares recorded times with the assigned schedule blocks, excludes configured breaks, and counts only work overlapping those blocks. Late and undertime minutes are recorded; payable hours are limited by scheduled time and the application's configured penalty rounding. Incomplete logs and days without a schedule do not receive payable hours.
6. Authorized payroll staff can review attendance logs, search and filter records, and add manual attendance for scheduled dates. Faculty accounts can view their own daily time record and attendance summary.
7. Payroll staff create semi-monthly payroll periods and generate draft payroll records for active faculty. The system derives hours and regular gross pay from attendance and the configured daily or hourly rate. Other earnings, adjustments, and deductions are entered or reviewed by payroll staff; generated records begin with these additional amounts set to zero.
8. Payroll records move through draft, approved, and released statuses. Draft records can be edited. Faculty members see only payslips released to their own account. Payslips can be printed or downloaded as A4 PDF files.

9. Administrative screens provide access to dashboard summaries, faculty records, attendance logs, payroll periods and records, account settings, rank rates, general settings, and database backup and restore functions.
10. The attendance firmware displays the current time and scan status. The TFT build can retain scans in local flash while offline and upload them when the connection returns. The application records hardware attendance events so device retries can be recognized. Enrollment status, cancellation, and credential replacement are coordinated between the web application and the device.

The system is intended to support an integrated workflow for attendance monitoring and payroll record preparation. Payroll calculations and generated payslips remain subject to review by authorized payroll personnel and to the rates and attendance information entered in the application.

The developed system may benefit the following:

1. COS Faculty Members. Faculty can record attendance using an assigned RFID card or registered fingerprint, review their own attendance information, and access payslips after payroll staff release them.
2. Department of Computer Studies. The department can maintain faculty, attendance, schedule, and payroll records in a centralized application and review information without manually retrieving device logs through USB.
3. Authorized Personnel. Administrators can maintain user and faculty records, while payroll staff can review attendance, create payroll periods, manage payroll details, approve records, and release payslips according to their assigned access.

4. Researchers. The project provides experience in web application development, relational data management, RFID and fingerprint integration, ESP32 firmware, offline event handling, payroll rules, and system evaluation.
5. Future Researchers. The system may serve as a reference for projects involving attendance monitoring, payroll record management, RFID, biometric matching, connected devices, and offline synchronization.

Time and Place of the Study

The study, titled "Payroll Management System for Contract of Service Faculty Members of the Cavite State University - Imus Campus Using RFID," is conducted for the Department of Computer Studies at Cavite State University - Imus Campus, City of Imus, Cavite. The stated study period is August 2026 to October 2026. Activities include gathering requirements, reviewing related literature, designing and developing the system, integrating its hardware and software, conducting tests, orienting intended users, gathering feedback, and making improvements. Actual implementation and evaluation results should be reported from the project's documented deployment and study activities.

Scope and Limitation of the Study

The study covers the development of a payroll management system for COS faculty members of the DCS at Cavite State University - Imus Campus. The implemented scope includes account access, faculty management, attendance monitoring, schedule configuration, payroll record processing, and payslip generation. The connected attendance device supports RFID and fingerprint capture. Authorized payroll staff can also enter attendance manually.

Account Module.

This module supports login, password recovery, email verification, and role-based access for administrators, payroll staff, and faculty. Administrative functions are protected by role checks. Faculty access is limited to their own attendance information and released payslips.

Faculty Management Module.

Authorized administrators can maintain faculty identity and contact details, employment classification and dates, rank and rate, service history, account association, schedule blocks, breaks, and attendance credentials. Credential registration is requested through the web system and completed by the connected device.

Attendance Monitoring Module.

This module records time-in and time-out from an active registered RFID card or an enrolled fingerprint identifier. RFID and fingerprint are both available identification methods in the current implementation. The application also provides manual attendance entry for authorized payroll staff. The device communicates with the server through a hardware API secured by a configured API key. The active TFT hardware uses an ESP32, MFRC522-compatible RFID reader, AS608 fingerprint sensor, DS3231 real-time clock, ILI9488 TFT, Wi-Fi, status indicators, and local flash storage.

Attendance records are stored centrally after server acceptance. When the device is offline or a temporary request failure occurs, it can preserve an event locally for retry using its event identifier and original timestamp. Uploads may arrive later than newer scans, so records should be reviewed when late synchronization affects the order of same-day events. Local queuing does not guarantee delivery if the device storage, clock, network, API configuration, or server is unavailable.

The application checks whether a faculty member has a schedule on the attendance date. The calculator uses assigned schedule blocks and breaks to determine payable hours, late minutes, undertime minutes, and attendance status. It does not count gaps between schedule blocks as payable time. The resulting log supports attendance review and payroll generation.

Payroll Module.

Payroll staff create semi-monthly periods and generate draft payroll records from attendance within the period. Regular pay uses recorded payable hours and the faculty member's configured hourly or daily rate. For daily-rate records, the system prorates pay against the scheduled hours for each attendance date. Attendance-based late and undertime effects are reflected in payable hours. Additional earnings and deductions are managed as payroll record fields and require staff review; the system does not automatically derive statutory contributions from external government tables.

Payroll records can include overtime pay, other earnings, increases, withholding tax, GSIS, PhilHealth, Pag-IBIG, loan deductions, FEA dues, OBA, and CRA fields. Newly generated records set these additional values to zero. Payroll staff can edit draft records, approve them, and release them. Faculty users can view their own released payslip, which can be printed or downloaded as a PDF. Payroll records are based on the attendance and rates in the application and require verification by authorized personnel.

Reports and Record Access.

Authorized users can review dashboard summaries, searchable and filterable attendance records, payroll periods, payroll records, and payslips. The system provides a printable and downloadable payslip PDF. It does not claim to replace university-wide accounting, finance, human resources, or official payroll systems; its records and outputs are limited to the implemented project scope.

The study is limited to COS faculty members under the Department of Computer Studies at the Imus Campus. It does not cover faculty from other departments, other employee classifications, or other CvSU campuses. The deployment environment, number of actual users, operational acceptance, and evaluation outcomes must be established by the study and are not inferred from the source code.

Attendance capture depends on properly assigned credentials, functioning sensors, a usable device clock, power, and correct configuration. Online server recording depends on Wi-Fi or internet availability and a working API endpoint and key. The TFT device can queue events during temporary connectivity failures, but queued records must eventually be accepted by the server. A new or replaced fingerprint sensor does not contain the old sensor's templates; faculty fingerprints must be enrolled on that physical sensor. The application stores fingerprint identifiers, not a biometric image or template for remote matching.

Payroll results depend on accurate faculty rates, schedules, attendance, and staff-entered additions or deductions. The software supports review and processing of these records but does not establish that a payroll record has been audited, paid, or accepted by the university. Hardware tests, user acceptance, deployment status, and measured performance should be reported from actual study evidence.

Conceptual Framework of the Study

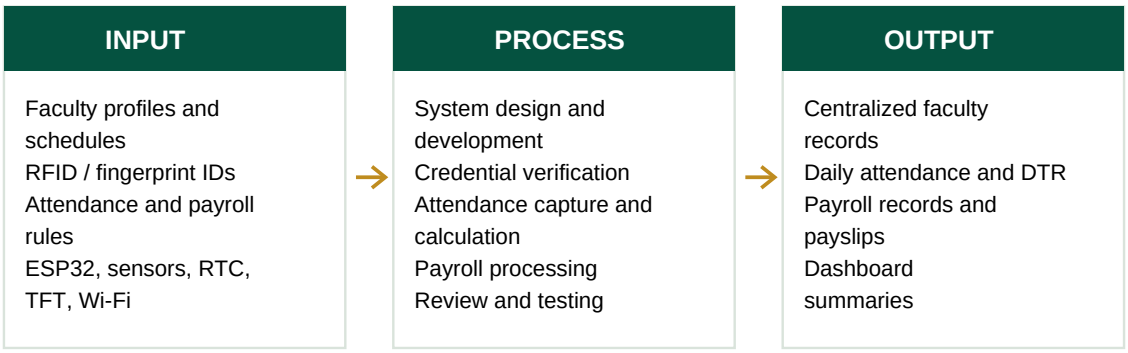


Figure 1: Conceptual Framework of the Study

Input.

Inputs include faculty identity and employment information, account roles, RFID card and fingerprint identifiers, assigned schedules and breaks, attendance events, faculty rates, payroll period dates, and payroll additions or deductions. Technology inputs include the web application and database, ESP32 attendance firmware, RFID reader, fingerprint sensor, real-time clock, TFT display, Wi-Fi connection, and supporting power and network infrastructure.

Process.

The process includes requirements analysis, system design, iterative development, faculty and schedule setup, credential enrollment, attendance capture, server validation, schedule-based attendance calculation, payroll generation, staff review and approval, payslip release, testing, and improvement. When the device cannot reach the server, it may retain attendance events and retry synchronization when connectivity returns.

Output.

The output is a web-based payroll management system for COS faculty members of the DCS at Cavite State University - Imus Campus. It provides centralized faculty and attendance records, daily time records, payroll period and payroll records, and printable or downloadable payslips. Attendance can be captured through RFID or fingerprint using the connected device, with manual entry available to authorized payroll staff. The system is designed to support review and processing; actual deployment and evaluation findings are determined by the study.

Definition of Terms

The following terms are defined according to their use in the current system:

AS608 Fingerprint Sensor.

A UART fingerprint module used by the ESP32 attendance device to capture fingerprint images and match them against templates stored in the sensor. The application receives the matched sensor slot identifier to associate the attendance event with a faculty credential.

Attendance Log.

A database record for a faculty member and attendance date that can contain time-in, time-out, identification methods, status, late and undertime minutes, payable hours, and remarks.

Centralized Database.

The application database that stores user, faculty, schedule, credential identifier, attendance, payroll, and related system records for access through the web application and hardware API.

Contract of Service (COS) Faculty Member.

A faculty member engaged under a Contract of Service and included in the DCS scope of this study.

DS3231 Real-Time Clock (RTC).

A clock module connected to the ESP32 that maintains time when internet access is unavailable. The active TFT firmware uses it to timestamp attendance scans and displays time in Philippine time.

ESP32.

A microcontroller used in the attendance hardware to read RFID cards and fingerprint sensor results, communicate with the web application over Wi-Fi, control the display and indicators, and retain attendance events for later upload when needed.

Fingerprint Credential Identifier.

The identifier associated with a fingerprint template slot in the physical sensor and stored by the application to link a successful match to a faculty record. The fingerprint template itself is held by the sensor rather than the web application.

Payroll Period.

A date range for grouping attendance and generating payroll records. The application validates semi-monthly cutoffs from day 1 to 15 or day 16 to the last day of the month.

Payroll Management System.

The Laravel and React web application developed in this study to manage faculty data, schedules, attendance, payroll records, account roles, and payslips for the intended COS faculty scope.

RC522 / MFRC522 RFID Reader.

An RFID reader module connected to the ESP32. It reads a card UID, which the system compares with an active registered RFID credential to identify a faculty member.

Radio Frequency Identification (RFID).

A technology that uses radio communication to identify a compatible card or tag. In this system, an assigned RFID card is one of the attendance identification methods supported by the hardware.

Role-Based Access.

Access control that grants application functions according to user roles, including administrator, payroll staff, and faculty. Faculty access to attendance and payroll records is limited to the user's own information, with payroll access restricted to released payslips.

TFT Display.

A 3.5-inch ILI9488 color screen used in the active hardware build to show the clock, attendance status, connectivity, and queued scan information.

Time-In and Time-Out.

The two attendance events recorded for a faculty member on a scheduled work date. Each event can be captured with RFID, fingerprint, or entered manually by authorized payroll staff.

Offline Attendance Queue.

Local device storage used by the TFT firmware to retain attendance events when a server request cannot be completed. The firmware retries queued events when online and removes an event only after a successful server response.

Schedule-Based Attendance Calculation.

The application process that compares recorded attendance times with assigned schedule blocks and breaks to determine late and undertime values, attendance status, and payable hours for payroll processing.

Web Application Programming Interface (API).

The application endpoints used by the ESP32 to submit attendance events and coordinate credential enrollment. Hardware requests are authenticated using a configured API key.